2.23 Partial Fractions

Worked Solutions

1)

$$\frac{2}{x(x^2+2)} \equiv \frac{A}{x} + \frac{Bx+C}{x^2+2},$$
$$2 = A(x^2+2) + (Bx+C)x.$$

Let $x = 0$: $2 = 2A$, so $A = 1$.

$$\text{Let } x = 1: \quad 2 = 3A + B + C, \quad \text{so } B + C = -1.$$

$$\text{Let } x = 2: \quad 2 = 6A + 4B + 2C, \quad \text{so } 4B + 2C = -4, \quad \therefore B = -1,\ C = 0.$$

Therefore

$$\frac{2}{x(x^2+2)} = \frac{1}{x} - \frac{x}{x^2+2}.$$

2)

$$\frac{3x^2 - x + 10}{(x+1)(x^2+1)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+1},$$
$$3x^2 - x + 10 = A(x^2+1) + (Bx+C)(x+1).$$

Let $x = 0$: $10 = A + C$.

$$\text{Let } x = 1: \quad 12 = 2A + 2B + 2C, \quad \text{so } 6 = A + B + C, \quad \therefore B = -4.$$

Let $x = 2$: $20 = 5A + 6B + 3C$, giving $A = 7$ and $C = 3$. Thus

$$\frac{3x^2 - x + 10}{(x+1)(x^2+1)} = \frac{7}{x+1} - \frac{4x-3}{x^2+1}.$$

3 a)

The term in the numerator of the fraction with denominator $x^2 + 2$ should be of the form $Bx + C$.

3 b)

$$\frac{3x^2 + 5x + 5}{(x+1)(x^2+2)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+2},$$
$$3x^2 + 5x + 5 = A(x^2+2) + (Bx+C)(x+1).$$

Let $x = 0$: $5 = 2A + C$.

$$\text{Let } x = 1: \quad 13 = 3A + 2B + 2C, \qquad \text{let } x = 2: \quad 27 = 6A + 6B + 3C.$$

Solving gives $A = 1$, $B = 2$ and $C = 3$. Hence

$$\frac{3x^2 + 5x + 5}{(x+1)(x^2+2)} = \frac{1}{x+1} + \frac{2x+3}{x^2+2}.$$

4 a)

$$\frac{x+5}{(x^2+1)(x+1)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+1},$$
$$x+5 = A(x^2+1) + (Bx+C)(x+1).$$

Let $x=0$: $5 = A + C$. Let $x=1$: $6 = 2A + 2B + 2C$. Let $x=2$: $7 = 5A + 6B + 3C$. Thus $A = 2$, $B = -2$ and $C = 3$, so

$$\frac{x+5}{(x^2+1)(x+1)} = \frac{2}{x+1} + \frac{-2x+3}{x^2+1} = \frac{2}{x+1} - \frac{2x-3}{x^2+1}.$$

4 b)

$$\frac{5x}{(2x+1)(x^2+1)} \equiv \frac{A}{2x+1} + \frac{Bx+C}{x^2+1},$$
$$5x = A(x^2+1) + (Bx+C)(2x+1).$$

Let $x=0$: $0 = A + C$. Let $x=1$: $5 = 2A + 3B + 3C$. Let $x=2$: $10 = 5A + 10B + 5C$. Thus $A = -2$, $B = 1$ and $C = 2$, so

$$\frac{5x}{(2x+1)(x^2+1)} = -\frac{2}{2x+1} + \frac{x+2}{x^2+1}.$$

4 c)

$$\frac{x-2}{(x+3)(x^2+4)} \equiv \frac{A}{x+3} + \frac{Bx+C}{x^2+4},$$
$$x-2 = A(x^2+4) + (Bx+C)(x+3).$$

Let $x=0$: $-2 = 4A + 3C$. Let $x=1$: $-1 = 5A + 4B + 4C$. Let $x=2$: $0 = 8A + 10B + 5C$. Thus $A = -\frac{5}{13}$, $B = \frac{5}{13}$ and $C = -\frac{2}{13}$, so

$$\frac{x-2}{(x+3)(x^2+4)} = -\frac{5}{13(x+3)} + \frac{5x-2}{13(x^2+4)}.$$

4 d)

$$\frac{3x^2-6x+15}{(2x^2+1)(x-1)} \equiv \frac{A}{x-1} + \frac{Bx+C}{2x^2+1},$$
$$3x^2-6x+15 = A(2x^2+1) + (Bx+C)(x-1).$$

Let $x=0$: $15 = A - C$. Let $x=1$: $12 = 3A$, so $A = 4$ and $C = -11$. Let $x=2$: $15 = 9A + 2B + C$, giving $B = -5$. Therefore

$$\frac{3x^2-6x+15}{(2x^2+1)(x-1)} = \frac{4}{x-1} - \frac{5x+11}{2x^2+1}.$$

4 e)

Let $f(x) = 2x^3 - 3x^2 + 8x - 12$, so

$$f\left(\frac{3}{2}\right) = \frac{27}{4} - \frac{27}{4} + 12 - 12 = 0,$$

therefore $(2x - 3)$ is a factor. Thus

$$2x^3 - 3x^2 + 8x - 12 = (2x - 3)(x^2 + 4).$$

$$\begin{aligned}\frac{25x^2}{2x^3 - 3x^2 + 8x - 12} &= \frac{25x^2}{(2x - 3)(x^2 + 4)} \\ &\equiv \frac{A}{2x - 3} + \frac{Bx + C}{x^2 + 4}, \\ 25x^2 &= A(x^2 + 4) + (Bx + C)(2x - 3).\end{aligned}$$

Let $x = 0$: $0 = 4A - 3C$. Let $x = 1$: $25 = 5A - B - C$. Let $x = 2$: $100 = 8A + 2B + C$. Thus $A = 9$, $B = 8$ and $C = 12$, so

$$\frac{25x^2}{2x^3 - 3x^2 + 8x - 12} = \frac{9}{2x - 3} + \frac{8x + 12}{x^2 + 4}.$$

4 f)

Let $f(x) = x^3 - 3x^2 + 5x - 15$, so $f(3) = 27 - 27 + 15 - 15 = 0$, therefore $(x - 3)$ is a factor. Hence

$$x^3 - 3x^2 + 5x - 15 = (x - 3)(x^2 + 5).$$

$$\begin{aligned}\frac{x^2}{x^3 - 3x^2 + 5x - 15} &\equiv \frac{A}{x - 3} + \frac{Bx + C}{x^2 + 5}, \\ x^2 &= A(x^2 + 5) + (Bx + C)(x - 3).\end{aligned}$$

Let $x = 0$: $0 = 5A - 3C$. Let $x = 1$: $1 = 6A - 2B - 2C$. Let $x = 2$: $4 = 9A - 2B - C$. Thus $A = \frac{9}{14}$, $B = \frac{5}{14}$ and $C = \frac{15}{14}$, so

$$\frac{x^2}{x^3 - 3x^2 + 5x - 15} = \frac{9}{14(x - 3)} + \frac{5x + 15}{14(x^2 + 5)}.$$

4 g)

$$\begin{aligned}\frac{1}{(x^2 + k)(x - 1)} &\equiv \frac{A}{x - 1} + \frac{Bx + C}{x^2 + k}, \\ 1 &= A(x^2 + k) + (Bx + C)(x - 1).\end{aligned}$$

Let $x = 0$: $1 = kA - C$. Let $x = 1$: $1 = (k + 1)A$, so $A = \frac{1}{k+1}$ and $C = \frac{k}{k+1} - 1 = -\frac{1}{k+1}$. Compare squared terms: $0 = A + B$. Therefore $A = \frac{1}{k+1}$, $B = -\frac{1}{k+1}$ and $C = -\frac{1}{k+1}$. Hence

$$\frac{1}{(x^2 + k)(x - 1)} = \frac{1}{(k + 1)(x - 1)} - \frac{x + 1}{(k + 1)(x^2 + k)}.$$

4 f)

$$\frac{k}{(x^2+k)(x-k)} \equiv \frac{A}{x-k} + \frac{Bx+C}{x^2+k},$$
$$1 = A(x^2+k) + (Bx+C)(x-k).$$

Let $x=0$: $1 = kA - kC$. Let $x=k$: $1=(k^2+k)A$, so $A = \frac{1}{k+1}$ and $C = \frac{k(1/(k+1))-1}{k} = -\frac{k}{k+1}$. Compare squared terms: $0 = A+B$. Therefore $A = \frac{1}{k+1}$, $B = -\frac{1}{k+1}$ and $C = -\frac{k}{k+1}$. Hence

$$\frac{k}{(x^2+k)(x-k)} = \frac{1}{(k+1)(x-k)} - \frac{x+k}{(k+1)(x^2+k)}.$$

5)

$$\frac{6x^2+5x+6}{x^3+kx} \equiv \frac{A}{x} + \frac{Bx+C}{x^2+k},$$
$$6x^2+5x+6 = A(x^2+k) + (Bx+C)x.$$

Let $x=0$: $6 = kA$, so $A = \frac{6}{k}$. Since A is a positive integer, $k = 1, 2, 3$ or 6.

Compare quadratic terms: $6 = A+B$, so $6 = \frac{6}{k} + B$, and $B = 6 - \frac{6}{k}$. Since B is a positive integer, $k = 2, 3$ or 6. Compare linear terms: $5 = C$. Therefore $k=2$, $k=3$ or $k=6$.

6 a)

$$\frac{18}{(x-1)(x+2)(x-4)} \equiv \frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{x-4},$$
$$18 = A(x+2)(x-4) + B(x-1)(x-4) + C(x-1)(x+2).$$

Let $x=1$: $18 = -9A$, so $A=-2$. Let $x=-2$: $18 = 18B$, so $B=1$. Let $x=4$: $18 = 18C$, so $C=1$. Therefore

$$\begin{aligned}\int \frac{18}{(x-1)(x+2)(x-4)}\,dx &= \int \left(-\frac{2}{x-1} + \frac{1}{x+2} + \frac{1}{x-4}\right) dx \\ &= -2\ln|x-1| + \ln|x+2| + \ln|x-4| + c \\ &= -\ln|(x-1)^2| + \ln|(x+2)(x-4)| + c \\ &= \ln\left|\frac{(x+2)(x-4)}{(x-1)^2}\right| + c.\end{aligned}$$

6 b)

$$\frac{x-2}{(x-1)^2} \equiv \frac{A}{x-1} + \frac{B}{(x-1)^2},$$
$$x-2 = A(x-1) + B.$$

Let $x=1$: $-1 = B$, so $B=-1$. Let $x=0$: $-2 = -A+B$, so $A=1$. Therefore

$$\int \frac{x-2}{(x-1)^2}\,dx = \int \left(\frac{1}{x-1} - \frac{1}{(x-1)^2}\right) dx$$

$$= \int \left(\frac{1}{x-1} - (x-1)^{-2} \right) dx$$
$$= \ln|x-1| + (x-1)^{-1} + c$$
$$= \ln|x-1| + \frac{1}{x-1} + c.$$

6 c)

$$\frac{7x}{(2x-1)(x+3)^2} \equiv \frac{A}{2x-1} + \frac{B}{x+3} + \frac{C}{(x+3)^2},$$
$$7x = A(x+3)^2 + B(2x-1)(x+3) + C(2x-1).$$

Let $x = \frac{1}{2}$: $\frac{7}{2} = \frac{49}{4}A$, so $A = \frac{2}{7}$. Let $x = -3$: $-21 = -7C$, so $C = 3$. Let $x = 0$: $0 = 9A - 3B - C$, so $B = -\frac{1}{7}$. Therefore

$$\int \frac{7x}{(2x-1)(x+3)^2}\, dx = \int \left(\frac{2}{7(2x-1)} - \frac{1}{7(x+3)} + \frac{3}{(x+3)^2} \right) dx$$
$$= \frac{1}{7}\ln|2x-1| - \frac{1}{7}\ln|x+3| - 3(x+3)^{-1} + c.$$

7)

$$\frac{3x^2+x+4}{(x+1)(x^2+2)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+2},$$
$$3x^2 + x + 4 = A(x^2+2) + (Bx+C)(x+1).$$

Let $x = 0$: $4 = 2A + C$. Let $x = 1$: $8 = 3A + 2B + 2C$. Let $x = 2$: $18 = 6A + 6B + 3C$. Thus $A = 2$, $B = 1$ and $C = 0$, so

$$\int \frac{3x^2+x+4}{(x+1)(x^2+2)}\, dx = \int \left(\frac{2}{x+1} + \frac{x}{x^2+2} \right) dx$$
$$= 2\ln|x+1| + \frac{1}{2}\ln|x^2+2| + c$$
$$= \ln|(x+1)^2| + \ln|\sqrt{x^2+2}| + c$$
$$= \ln\left((x+1)^2\sqrt{x^2+2} \right) + c, \quad \text{as required.}$$

8)

$$\frac{x^2-4x+15}{(x+1)(x^2+3)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+3},$$
$$x^2 - 4x + 15 = A(x^2+3) + (Bx+C)(x+1).$$

Let $x = 0$: $15 = 3A + C$. Let $x = 1$: $12 = 4A + 2B + 2C$. Let $x = 2$: $11 = 7A + 6B + 3C$. Thus $A = 5$, $B = -4$ and $C = 0$, so

$$\frac{x^2-4x+15}{(x+1)(x^2+3)} = \frac{5}{x+1} - \frac{4x}{x^2+3}.$$

$$\begin{aligned}\int_1^4 \frac{x^2-4x+15}{(x+1)(x^2+3)}\,dx &= \int_1^4 \left(\frac{5}{x+1}-\frac{4x}{x^2+3}\right)dx\\ &= \left[5\ln|x+1|-2\ln|x^2+3|\right]_1^4\\ &= (5\ln 5-2\ln 19)-(5\ln 2-2\ln 4)\\ &= \ln\frac{5^5}{19^2}-\ln\frac{2^5}{4^2}\\ &= \ln\frac{3125}{361}-\ln 2 = \ln\frac{3125}{722}, \quad \text{as required.}\end{aligned}$$

9)

$$\begin{aligned}\frac{4x^2+6x+2}{(x+2)(x^2+2)} &\equiv \frac{A}{x+2}+\frac{Bx+C}{x^2+2},\\ 4x^2+6x+2 &= A(x^2+2)+(Bx+C)(x+2).\end{aligned}$$

Let $x=0$: $2=2A+2C$. Let $x=1$: $12=3A+3B+3C$. Let $x=2$: $30=6A+8B+4C$. Thus $A=1$, $B=3$ and $C=0$, so

$$\frac{4x^2+6x+2}{(x+2)(x^2+2)} = \frac{1}{x+2}+\frac{3x}{x^2+2}.$$

$$\begin{aligned}\int \frac{4x^2+6x+2}{(x+2)(x^2+2)}\,dx &= \int \left(\frac{1}{x+2}+\frac{3x}{x^2+2}\right)dx\\ &= \ln|x+2|+\frac{3}{2}\ln|x^2+2|+c\\ &= \ln\left|(x+2)(x^2+2)^{3/2}\right|+c.\end{aligned}$$

When $y=0$, $4x^2+6x+2=0$, so $x=-1$ or $x=-\frac{1}{2}$. The shaded-area limits are $x=-\frac{1}{2}$ to $x=0$:

$$\begin{aligned}\int_{-1/2}^0 \frac{4x^2+6x+2}{(x+2)(x^2+2)}\,dx &= \left[\ln\left|(x+2)(x^2+2)^{3/2}\right|\right]_{-1/2}^0\\ &= \ln|4\sqrt{2}|-\ln\left|\frac{81}{16}\right|\\ &= \ln\left(\frac{16\times 4\sqrt{2}}{81}\right) = \ln\frac{64\sqrt{2}}{81}.\end{aligned}$$

10)

$$\begin{aligned}\frac{(k-1)x^2-3kx-4}{(x-3)(x^2+4)} &\equiv \frac{A}{x-3}+\frac{Bx+C}{x^2+4},\\ (k-1)x^2-3kx-4 &= A(x^2+4)+(Bx+C)(x-3).\end{aligned}$$

Let $x=0$: $-4=4A-3C$. Compare linear terms: $-3k=C-3B$. Compare quadratic terms: $k-1=A+B$. Thus $A=-1$, $B=k$ and $C=0$, and

$$\frac{(k-1)x^2-3kx-4}{(x-3)(x^2+4)} = -\frac{1}{x-3}+\frac{kx}{x^2+4}.$$

$$\begin{aligned}\int \frac{(k-1)x^2-3kx-4}{(x-3)(x^2+4)}\,dx &= \int\left(\frac{kx}{x^2+4}-\frac{1}{x-3}\right)dx\\ &= \frac{k}{2}\ln|x^2+4|-\ln|x-3|+c\\ &= \ln\left|\frac{(x^2+4)^{k/2}}{x-3}\right|+c.\end{aligned}$$

$$\begin{aligned}\int_4^6 \frac{(k-1)x^2-3kx-4}{(x-3)(x^2+4)}\,dx &= \left[\ln\left|\frac{(x^2+4)^{k/2}}{x-3}\right|\right]_4^6\\ &= \ln\left|\frac{40^{k/2}}{3}\right|-\ln|20^{k/2}|\\ &= \ln\left(\frac{40^{k/2}}{3\times 20^{k/2}}\right).\end{aligned}$$

Compare with the given $\ln(32/3)$:

$$\frac{40^{k/2}}{20^{k/2}}=32 \quad\Rightarrow\quad \left(\frac{40}{20}\right)^{k/2}=32 \quad\Rightarrow\quad 2^{k/2}=32 \quad\Rightarrow\quad k=10.$$